Presence Without Performance:

Mechanism-Based Guidance for Supporting Cancer Patients and Caregivers

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Companion paper to McCarthy 2026, Paper 8: "The Cost of Computation"

Abstract

A 2009 study put rats with mammary tumor predisposition into either group cages or solo cages and waited. The socially isolated rats developed mammary tumor burden 84 times higher than the group-housed controls. The molecular pathway is now mapped: corticoamygdala circuit → β-adrenergic and HPA outflow → tumor microenvironment. The same mechanism shows up in human meta-analyses of β-blocker use, shift work, depression, and obstructive sleep apnea, all with effect sizes large enough to be clinically meaningful.

This paper argues two things. First, treatment of depression, isolation, and chronic stress in cancer patients is not adjunct care — it is part of cancer treatment, because it intervenes on the same molecular pathways as the disease. Second — and harder — the *kind* of social support that helps is more specific than the literature usually says. The mouse data dissociates two things conventional support advice tends to fuse: physical presence and relational performance. Mice cannot perform emotional labor for each other. The 84× effect comes from cohabitation alone. When we translate this to human patients, the active ingredient is **presence**, not performance. Support that adds emotional accounting work to the patient — even kind, well-meaning support — is iatrogenic by the same molecular pathway it is meant to protect.

This paper provides specific guidance, with mechanism, for what kind of support to provide cancer patients and caregivers, and which familiar forms of support are quietly counter-therapeutic.

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1. The cage and the room

A rat in a cage with other rats does not get cancer at the rate a rat alone in a cage gets cancer. The difference is large. In Hermes et al. 2009, rats genetically predisposed to mammary tumors were assigned to either group housing or solo housing and followed. The solo-housed animals developed mammary tumor burden 84 times higher than the group-housed animals. The relative risk for invasive ductal carcinoma was 3.3 [Hermes et al., *PNAS* 2009].

The rats in the group cage are not having conversations. They are not asking each other how their day is going. They are not expressing concern. They are not sharing feelings. They are just there. Their bodies are in proximity to other bodies of the same species. That is the entire intervention. And the intervention is large enough that you would call it a major drug effect if it were a pill.

A patient in a hospital room is in a different category from a rat in a cage in important ways. But the molecular machinery that processes social presence is conserved across mammals. The chronic-stress / cancer literature in humans, when read alongside the rodent housing literature, points at the same axis with the same direction. People with major depression have 23% to 83% higher cancer mortality across five common cancer types after adjustment for clinical factors [meta-analysis of 65 studies, *GeroScience* 2025]. People who do night shift work have elevated breast, prostate, and colon cancer risk; the IARC classifies night shift work as Group 2A (probably carcinogenic to humans) [IARC Monograph Vol. 124]. People taking β-blockers chronically — drugs that block sympathetic nervous system signaling at peripheral tissues — have reduced cancer mortality in melanoma and breast cancer in observational studies [Choy et al., 2016; multiple meta-analyses].

These findings are not about willpower or attitude. They are about the same molecular pathway that responds to whether a rat is alone or with other rats: β-adrenergic signaling on tumor stroma, HPA axis modulation of immune surveillance, glucocorticoid effects on cell-cycle regulation, vagal tone modulating inflammation. The pathway is documented in detail. The corticoamygdala circuit that mediates the rodent social-housing effect was identified at single-circuit resolution in *Neuron* in 2025 [Neuron 2025]. **This is biology, not metaphor.**

What we want to do in this paper is take that biology seriously and follow it where it leads, which is into a specific and somewhat uncomfortable revision of how friends, families, and clinicians support cancer patients. Some of the conventional wisdom is correct. Some of it actively makes outcomes worse, by the same molecular pathway it is meant to protect.

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2. The convergence: three independent scales of evidence

The biology supporting the claim "social context modulates cancer outcomes through documented molecular pathways" comes from three different scales, each measured by different methods, each pointing the same direction.

2.1 Within-individual modulation in humans

The within-individual evidence is the largest body of work. Specific lesion conditions — natural and pharmacologic — that disrupt brain-body coupling produce measurable changes in cancer outcomes:

- **Major depression** is associated with 23–83% higher cancer mortality across colorectal, breast, lung, prostate, and pancreatic cancers in a 2025 meta-analysis of 65 cohort studies. The effect persists after adjustment for clinical confounders, leading the authors to suggest depression "may play a causal role" rather than being purely a marker of disease severity.
- **Insomnia and obstructive sleep apnea** show elevated cancer prevalence in meta-analyses spanning more than 32 million patients. The lung cancer hazard ratio in OSA patients with long follow-up is 1.28. The proposed mechanism is intermittent hypoxia plus circadian disruption.
- **Shift work** is classified Group 2A by IARC ("probably carcinogenic to humans") on the basis of evidence for breast, prostate, colon, and rectal cancers. The mechanism involves circadian disruption, melatonin suppression, and disruption of the glymphatic / immune coordination cycle.
- **Chronic β-blocker use**, in the opposite direction, is associated with reduced cancer mortality in melanoma and breast cancer in multiple observational studies. Recent rigorous meta-analyses have softened some of the original signal (immortal time bias may have inflated earlier estimates), but the direction is consistent: blocking sympathetic nervous system

signaling reduces cancer mortality, with the strongest evidence in cancers where the tumor microenvironment is β-adrenergic-responsive.

- **Vagotomy** (the surgical severing of the vagus nerve, performed historically for ulcer disease) is associated with altered cancer rates in vagally-innervated organs, with the direction depending on cancer type — supporting the broader claim that brain-body autonomic coupling matters for cancer biology, though the pattern is not uniformly directional.

These are not separate phenomena. They are the same axis (autonomic / circadian / stress regulation of cancer microenvironment) measured at the human within-individual scale by different methods. Each is published as an isolated finding. The framework presented here unifies them.

2.2 Within-species controlled experiments

Where the within-individual human evidence is observational and confounded, the rodent housing experiments are controlled. Same genome, same diet, same cage type — only the social condition varies.

- **Hermes et al. 2009 (PNAS)**: In socially isolated rats, mammary tumor burden was 84 times higher than in group-housed controls, with 3.3× relative risk for ductal carcinoma in situ and invasive ductal carcinoma. Isolated animals showed dysregulated corticosterone responses to everyday stressors, manifesting in young adulthood, *months before tumor development*. This is not stress inflicted by sickness — it is a baseline endocrine state in isolation that precedes and predicts the cancer.
- **Williams et al. 2009 (Cancer Prevention Research)**: Earlier confirmation of the same axis, leading to the widely-reported "social isolation worsens cancer" finding.
- **Madden et al. 2013 (Brain Behavior Immunity)**: Adverse social environment alters mammary gland gene expression and accelerates tumor growth in immunocompetent mouse strains.
- **Rats with dormant tumors (MDPI Cells 2023)**: Of recurring tumors, 45% recurred in group-housed animals; 75% recurred in socially isolated animals. Social isolation activated dormant mammary tumors and modified inflammatory and mitochondrial metabolic pathways in the mammary gland.
- **AACR Cancer Research 2022 (Abstract P5-01-12)**: Under thermoneutral housing temperatures (eliminating the cold-stress confound that has plagued mouse cancer studies), 4T1 mammary tumors grew significantly larger in physically and visually isolated mice compared to group-housed controls.
- **Neuron 2025**: A specific neural circuit — the corticoamygdala pathway — was identified as the mediator by which social interaction in mice suppresses breast cancer progression. This is the cleanest molecular bridge yet between social environment and tumor microenvironment.

Note what these experiments cannot test: the *content* of the social interaction. Mice and rats cannot ask each other for emotional support. They cannot reciprocate. They cannot demand updates or expect gratitude. The variable is purely physical cohabitation. **The 84× effect is the effect of being in the room with another mammal of the same species, with no other content required.** This will matter for the clinical translation in §4.

2.3 Cross-species (under construction)

If the within-individual and within-species evidence are correct, then across mammalian species we should see: animals whose ecology produces sustained low-baseline autonomic activation (apex predators, megafauna without natural predators, refuge species, deep-herd species) should show lower cancer rates than animals whose ecology produces sustained high-baseline autonomic activation (small prey, solitary vigilant species), after controlling for body mass and lifespan.

The cross-species data to test this directly is still limited. The two best published cross-mammalian cancer datasets — Boddy et al. 2020 (37 species) and Vincze et al. 2022 (192 vertebrates, ~34 mammals in the cleaned subset) — both lack the framework's strongest corner cases (naked mole rat, cetaceans, mice, voles). Within the species they do contain, a regression analysis with body mass, lifespan, mass-specific metabolic rate, and body temperature yields one robust finding — **lifespan is significantly negatively correlated with cancer prevalence after controlling for body mass** — and one marginal finding — **body temperature is marginally negatively correlated** at $p \approx 0.075$. The substrate-hardening axis is partially visible in the data we have but the test is statistically underpowered for the corner cases that would make it definitive. We report this honestly. The cross-species projection is the weakest of the three scales at present and is not where the case rests.

2.4 What the convergence rules out

What makes the case is not the strength of any single projection. It is the rank order and direction of multiple projections at different scales, all consistent with the same axis. Table 1 summarizes.

Scale	Method	Effect direction	Status
Within-individual (human β-blocker)	Pharmacologic / observational meta-analysis	sympathetic blockade $\downarrow$ cancer mortality	replicated, with caveats
Within-individual (human shift work)	Occupational epidemiology + IARC review	circadian disruption $\uparrow$ cancer	IARC Group 2A
Within-individual (human OSA)	Clinical cohort meta-analysis	sleep fragmentation $\uparrow$ cancer	replicated
Within-individual (human depression)	65-cohort meta-analysis	depression $\uparrow$ cancer mortality 23-83%	replicated, persists after adjustment
Within-species (rodent housing)	Controlled experiment	isolation 84$\times$ $\uparrow$ tumor burden	replicated, mechanism mapped
Within-species (rodent dormant tumor reactivation)	Controlled experiment	isolation reactivates dormant tumors	replicated 2023
Within-species (mouse, neural circuit)	Optogenetic / circuit dissection	corticoamygdala mediates social$\rightarrow$tumor effect	identified 2025
Cross-species (mammals, body temp)	Phylogenetic regression	lower Tb trend $\downarrow$ cancer	marginal at $p \approx 0.075$

Cross-species (mammals, lifespan)	Phylogenetic regression	longer lifespan ↓ cancer per unit time	significant at $p \approx 0.02$

The rule of an alternative explanation that could account for all rows of Table 1 is: there is no such alternative. Every plausible alternative account explains some rows but contradicts others. The autonomic / circadian / chronic-stress modulation axis is the unique frame consistent with the rank order and direction of all of them.

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3. The molecular bridge

The molecular pathway connecting social context to cancer microenvironment is the same one that connects other forms of "context overload" to cancer. McCarthy 2026 (Paper 8, "The Cost of Computation") identifies the upstream substrate: chromatin remodeling genes (ARID1A, KMT2C/D, SMARCA4, DNMT3A) are simultaneously the molecular implementation of context switching during cognition AND the most frequently mutated gene class in cancer. The same machinery that enables cells to switch which genes they express is the machinery whose somatic mutation drives uncontrolled gene expression in tumors. CHIP (clonal hematopoiesis of indeterminate potential), where 80% of mutations are in DNMT3A, TET2, and ASXL1, is the strongest direct evidence for this — these are exactly the chromatin and methylation machinery genes, and they fail first because they are the most actively read/written substrate in the genome.

The relevance to social context is that **chronic sympathetic / HPA activation upregulates exactly this machinery**. β-adrenergic signaling on tumor stromal cells modulates inflammatory cytokine production, immune evasion, angiogenesis, and metastasis. Chronic glucocorticoid exposure modifies chromatin accessibility patterns through specific GR (glucocorticoid receptor) binding sites. Sleep disruption interferes with the circadian regulation of DNA repair. Each of these is a documented molecular pathway that connects an external "context overload" condition to the same chromatin / epigenetic machinery that Paper 8 identifies as the shared substrate of cognition and cancer.

This means: when we tell a cancer patient to "manage stress" or "join a support group," what we are actually asking is for an intervention that will down-regulate β-adrenergic, HPA, and circadian signaling at the level of molecular signaling on the tumor microenvironment. The recommendation has a mechanism. It is not about feelings.

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4. The active ingredient is presence

This is the pivot point of the paper, and the part that requires the most care.

Look at what the rodent experiments measure. They measure tumor burden as a function of one variable: whether the animal is in a cage with other animals or alone. They do not measure the *quality* of the relationships between the animals. They cannot, because mice and rats do not have relationships in the sense humans use the word. They have proximity, ambient social cues (smell, sound, periodic contact), and the absence of the chronic vigilance state that solitary housing imposes on a normally social rodent. **And that is enough to produce an 84× effect on tumor burden.**

This dissociates two things that human cancer support literature usually fuses:

1. **Physical / contextual presence** — the active ingredient. Reduces baseline sympathetic tone purely by removing isolation. Does not require any emotional content from the patient.
2. **Relational performance** — emotional reciprocity, conversation, gratitude, narrating one's feelings, updating loved ones, maintaining the relationship. This is what "support" usually means in human contexts but is **not what the mouse data validates**.

The rodent data identifies (1) as the molecular intervention. (2) is what most well-meaning human supporters provide. Sometimes (1) and (2) come together (a friend sitting in the room reading their own book). Sometimes (2) comes without (1) (a phone call asking how you're feeling). The framework's claim is that (1) is the medical intervention; (2) is a separate thing that may or may not also be helpful, and which can actively backfire if it adds context-load to a patient whose substrate is already failing under context overload.

This is uncomfortable because it sounds like a critique of caring. It is not. It is a refinement of caring, in the direction the molecular biology actually points. Most supporters are providing (1) and (2) together, and the (1) part is doing the work. When supporters provide (2) without (1) — phone calls, texts, asking for updates from a distance, demanding emotional reciprocity, expecting thanks — the patient gets the cost of the relationship without the molecular benefit of presence.

Worse: the *demand* for emotional reciprocity in (2) imposes the same kind of context-load that the chronic-stress condition imposes in the first place. A patient who has to manage their family's feelings about their illness is paying a cognitive and emotional cost on top of the disease. By the framework, that cost upregulates the same pathways the support was meant to down-regulate. **It is iatrogenic by mechanism.**

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5. Wrong support and right support

The following table is the practical translation. It is the part of this paper that goes on a printable card for cancer support volunteers, family members of patients, and clinicians who counsel them.

Wrong support (adds context-load)	Right support (reduces context-load)
"Call us when you need anything." (Puts the burden of asking on the patient.)	"I'm coming over Tuesday at 4. Don't do anything. I'll let myself in."
Visits that require the patient to entertain, narrate, or reciprocate.	Visits that ask nothing of the patient.
"How are you feeling? Tell us everything."	Sitting quietly in the same room. Reading. Existing.
Help conditional on the patient's emotional performance ("It would mean a lot if you'd update us").	Unconditional presence. No scoreboard.
Support groups that mandate sharing as a requirement of participation.	Groups where attendance alone is enough.
Caregivers who need to be thanked.	Helpers who treat their help as theirs to give, with no ledger.
Friends who get hurt when they aren't kept informed.	Friends who hold space without expecting updates.

Wrong support (adds context-load)	Right support (reduces context-load)
"Let me know how I can help."	Bringing dinner without asking. Watering the plants without asking. Walking the dog without asking.
Performative concern: "I've been so worried about you."	Practical, low-demand action that the patient does not need to acknowledge.
Phone calls that require emotional energy from the patient to manage.	Texts that explicitly say "no need to respond".
Decision-deferring help: "I'll make lunch — what is everyone having?" (Real action, but the helper still requires the caregiver to make the decision for them.)	**Autonomous imperfect help**: "I'm making sandwiches for the kids. Whatever I find. You don't have to think about it." (Decision is made by the helper, even if imperfectly.)
The visiting parent who folds the laundry but second-guesses every other choice and routes the decisions back to the caregiver.	The visiting parent who walks in, picks the task, executes it, and leaves.

The right column is the operationalization of "presence without performance." Each row of the right column has the property that the patient is **not made to do anything**. Their context window is not loaded. Their emotional accounting work is not increased. The supporter is providing a low-demand, high-presence intervention that resembles, in the terms the biology cares about, what the rodent group cage provides: the presence of another conspecific without the demand of emotional performance.

This is harder than it sounds. Most well-meaning supporters are conditional in subtle ways they do not notice. The conditions are usually:

- They want to feel useful, so they require the patient to receive their help in a specific form
- They want their concern acknowledged, so they require updates and gratitude
- They want emotional connection, so they require the patient to share feelings
- They want clarity on what is needed, so they require the patient to identify and articulate their needs
- They want the relationship to feel reciprocal, so they require the patient to maintain their side
- They want to do the help correctly, so they require the caregiver to direct them at every decision point

The last item deserves its own name because it is the form of well-meaning support that family caregivers report as the most exhausting and the one they feel guilty complaining about. We call it **decision-deferring help**: real practical action — folding laundry, cooking the specific dinner that was requested, taking the kids somewhere — bundled with a chronic dependence on the caregiver to make every micro-decision. *"What is everyone having for lunch?" "Where do you keep the towels?" "Should I use this pan or that one?" "Is it okay if I take them to the park first?"* Each question is small. Each question is reasonable. Each question is, for someone in chronic-stress / context-overload state, a context-cost tax on attention they cannot afford to spend.

The discount can be severe enough to flip the sign of the help. A visiting helper who folds the laundry but routes ten decisions back to the caregiver in the process has not reduced the caregiver's context-load — they have added to it. The caregiver would have been better off folding the laundry alone, because the laundry is not the work; **the work is allocating attention and making decisions**, and the help that arrives is supposed to reduce that, not redirect it through a different channel.

The fix is **autonomous imperfect decisions over correct supervised ones**: a sandwich made wrong without asking is better than a sandwich made right with asking, because the asking is the cost. A helper who walks in, picks a task, executes it imperfectly, and leaves has provided more value than a helper who executes the same task perfectly while pulling the caregiver into every choice. The molecular pathway does not care whether the sandwich was the right kind. It cares whether the caregiver had to allocate cognitive budget to the helper's presence.

This is the part of the framework that is hardest for well-meaning supporters to internalize, because it asks them to be okay with making imperfect decisions in someone else's house, on behalf of someone else's family, often in domains where they feel unqualified. The framework's response is: **the imperfection is the help**. Asking a question to do it right delivers the question. Doing it wrong without the question delivers the help.

Each of these is a context demand on someone whose substrate is already failing under context overload. The framework reframes them as iatrogenic: well-meaning supporters who add context cost are increasing the patient's β-adrenergic and cortisol baseline and degrading the cancer-control substrate, even though their intent was the opposite.

Most adults have been socialized into reciprocal-relationship support models. Switching to non-reciprocal support feels emotionally wrong — like the supporter is being self-erasing or giving with strings attached. **The framework's claim is that for cancer patients (and caregivers, who are themselves in chronic-stress states), the non-reciprocal mode is medically required, not just emotionally generous.**

The mouse data is the load-bearing experimental anchor for that claim. Mice cannot perform emotional labor for each other. The 84× effect comes from cohabitation alone. Whatever it is in social presence that is the active ingredient, it does not require relational content. The conventional advice to "be there for someone" is pointing in the right direction. The conventional advice to "ask how they're doing, listen carefully, and let them lean on you" mixes the active ingredient with a context-cost overhead that, by the same biology, undoes part of the benefit.

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6. Caregivers also need non-reciprocal support

Everything in §5 also applies to the patient's primary caregivers. Caregivers of cancer patients live in chronic-stress states with measurable elevation of their own cortisol baseline, their own cardiovascular and cancer risk, and their own cognitive load. The caregiver is, biologically, also a patient — a patient at the early end of the same chronic-stress / substrate-degradation curve that the cancer patient is further down.

The problem for caregivers is that the support they need is doubly difficult to receive:

1. Caregivers are usually the ones organizing all the support around the patient. They are the ones answering the phone calls, scheduling the visits, managing the relatives. The infrastructure of "support for the patient" that violates the §5 right column lands on the caregiver, not on the patient. **Most well-meaning support of a cancer patient is, in practice, a context-load tax on the caregiver.**

2. Caregivers are socialized to be "the strong one" or "the one keeping it together." Performative strength is itself a context cost: the caregiver is not just managing the disease, they are managing their own emotional presentation to the people around them. By the framework, this is a chronic load that contributes to the caregiver's own elevated risk.

The right support for caregivers, structurally identical to the right support for patients, is **non-reciprocal presence**. Specifically:

- People who help the caregiver without asking the caregiver to coordinate the help
- Visitors who do not require the caregiver to be performing strength or composure
- Friends who do not need to be told how the patient is doing in order to feel involved
- Family members who do not require the caregiver to mediate their feelings about the patient's illness
- Practical help that arrives at the door, rather than help that requires the caregiver to scope, schedule, or accept

This is also harder than it sounds, for the same reasons.

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7. What this paper is not saying

The translational claim in this paper is uncomfortable in a specific way: it can be misread as victim-blaming or as endorsing magical-thinking interpretations of cancer that the field has rightly rejected. We need to be explicit about the distinction.

This paper says: - Chronic stress, isolation, and depression modulate cancer outcomes via documented β-adrenergic, corticoamygdala, and HPA pathways with effect sizes large enough to be clinically meaningful. - Treating depression and isolation in cancer patients is part of treatment because it intervenes on these pathways. - The active ingredient in beneficial social support is physical presence, not relational performance. - Supporters who add emotional accounting work to patients can, by the same molecular pathway, undo part of the benefit they intend to provide. - Caregivers need the same non-reciprocal support model because they are also in chronic-stress states.

This paper does not say: - Cancer patients caused their cancer by being sad, isolated, or stressed. The pathways modulate outcomes; they do not assign cause. Cancer has many drivers and the framework's claim is about modulation of an existing biological process, not about psychogenic origin. - Positive thinking cures cancer. The framework's predictions are about specific molecular interventions (β-adrenergic, HPA, circadian), not about cognitive or attitudinal states in any general sense. - All cancer is psychosomatic. Most of cancer biology is unrelated to the chronic-stress axis discussed here. The framework speaks to one component of the modulating environment, not to the whole disease. - Support groups are wrong, or that emotional support is harmful. The right kind of support — non-reciprocal presence — is medically beneficial. The wrong kind — context-loading reciprocity — is what the framework warns about. The book is open on which forms of support fall in which column for any given person; this paper provides the framework for asking the question.

The first set of claims is what the data supports. The second set of claims is what the field has been understandably allergic to. **The framework needs to plant its flag firmly in the first set and explicitly distance itself from the second**, because the failure mode (being misread as woo) would discredit the genuine biology that the data supports.

8. Conclusion

A rat in a cage with other rats does not get cancer at the rate a rat alone in a cage gets cancer, and the difference is 84×. The molecular pathway that produces that difference is conserved in humans. The within-individual epidemiology, the within-species controlled experiments, and the cross-species ecological signal all point at the same axis. The active ingredient is physical presence; the conventional advice that mixes presence with reciprocal performance is providing the active ingredient at a partial discount.

Cancer treatment programs that do not include presence-based support — and that do not actively distinguish it from performance-based support — are leaving treatment on the table. Cancer patients whose family and friends provide context-loading "help" are being treated worse than they would be if the same people simply showed up and read a book in the same room.

This is uncomfortable to read. The discomfort is partly because the recommendation is harder to follow than it sounds, and partly because it asks us to revise the conventional grammar of caring. But the biology is documented, the mechanism is mapped, the effect sizes are large, and the recommendation is concrete. **Just be there. Don't ask anything. Don't keep score. The molecular pathway that protects the patient does not need anything more than that, and it is degraded by anything more than that.**

For male patients in particular — who are statistically less likely to accept psychiatric or psychological intervention as part of their cancer treatment — the framework gives biological permission to take therapy, antidepressants, and quiet companionship as seriously as they take chemotherapy and radiation. None of these are softer interventions. They are upstream interventions on the same pathways the disease lives in.

For caregivers — who are also patients, at the early end of the same curve — the framework gives biological permission to receive non-reciprocal support without guilt. The help that arrives without asking, that requires nothing in return, is not selfishly accepted. It is medically necessary.

The conclusion that we want the reader to arrive at by the time they finish this paper is not "stress causes cancer." It is something more specific and more actionable: **the biology of cancer treatment includes the social environment of the patient, the social environment is best modulated by presence rather than performance, and the people around the patient need to know which kind of support they are providing.** Knowing the difference, and choosing presence, is part of treatment.

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(Not exhaustive — load-bearing citations only.)

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Status: Draft v1. Checkpoint. Not for distribution. Refinement targets: §2.3 (cross-species evidence) needs the regression result table once the data is unblocked; §3 may need an additional figure showing the molecular pathway; §5 table may benefit from a clinician-reviewed version; references list needs full bibliographic details.